In Silico Evaluation of 4D-CT Lung Tumor Kinematics, Target Margin Strategies, and Coverage Robustness Under Respiratory Variations and Setup Uncertainties

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ABSTRACT

Background: Intra-fraction respiratory motion causes pulmonary tumors to undergo substantial 3D excursions, complicating stereotactic body radiation therapy (SBRT). This study presents an open-source in silico framework evaluating 4D-CT kinematics, internal target volume (ITV) expansions, and anisotropic margins under unseen respiration and multi-regimen setup uncertainties (SBRT 3-fx, SBRT 5-fx, CFRT 30-fx).
Methods: An anthropomorphic thorax phantom was generated with an asymmetric 60:40 breathing cycle (T0 = Peak Inhalation; T0->T6 = 60% exhalation; T6 = End-Exhalation reference; T6->T0 = 40% active inhalation) and 36-degree AP/LR hysteresis. Three lesion sizes (r = 8, 12, 16 mm) and three breathing profiles were modeled. Frozen planning PTVs (Isotropic 2-12 mm, ITV + 0-5 mm, Anisotropic Mid-pos 2-4 mm) were tested across N=5,000 independent Monte Carlo cycles (exact 95% Wilson score CIs) and evaluated under combined breathing and setup uncertainties across 50 virtual patients for SBRT 3-fx, SBRT 5-fx, and CFRT 30-fx regimens (2 mm systematic, 2 mm random error) aligned with AAPM TG-324 Report 91.b.
Results: Centroid tracking achieved 3D MAE of 0.058 mm (r=8mm), 0.034 mm (r=12mm), and 0.021 mm (r=16mm). On planning 4D-CT, conventional 5 mm isotropic expansion achieved only 64.7% minimum coverage. Both ITV + 3 mm (23.9 cc) and Anisotropic + 2 mm (19.3 cc, -6.3% volume vs. 5 mm isotropic baseline) achieved >=99.4% planning coverage. However, under unseen irregular breathing with drift (N=5,000), ITV + 3 mm pass rate dropped to 87.1% (95% Wilson CI: [86.1%, 88.0%]; mean min CR: 97.6%, empirical range: [86.5%, 100.0%]), whereas ITV + 5 mm maintained a 97.9% pass rate (95% Wilson CI: [97.4%, 98.2%]). Under setup uncertainty, tight margins proved severely inadequate in SBRT: for 5-fraction SBRT, ITV + 3 mm yielded only 12.0% patient all-fraction success, whereas ITV + 5 mm restored all-fraction success to 62.0% (and 82.0% in 3-fraction SBRT). In 30-fraction conventional therapy, ITV + 5 mm achieved an 87.4% fraction pass rate and 70.0% course-averaged success.
Conclusions: While anisotropic and tight ITV margins provide geometric sparing under idealized regular breathing, they exhibit substantial coverage degradation under setup errors and drift. In concordance with AAPM TG-324, an ITV setup margin of 4-5 mm is clinically required for free-breathing lung SBRT unless daily volumetric IGRT (CBCT) and active motion management are deployed.

Keywords: *4D-CT, Respiratory Motion Modeling, Internal Target Volume (ITV), Setup Uncertainty, SBRT Fractionation, AAPM TG-324, Monte Carlo Simulation.*

1. Introduction

Radiation therapy for non-small cell lung cancer (NSCLC) presents a delicate balance between maximizing tumoricidal dose delivery and sparing surrounding functional lung parenchyma, cardiac structures, and the spinal cord [1]. Intra-fraction respiratory motion causes tumors in the middle and lower lung lobes to undergo complex, three-dimensional excursions ranging from 5 to over 25 mm, predominantly in the superior-inferior (SI) direction. Conventional planning relying on static 3D-CT scans captures an arbitrary snapshot of the respiratory cycle, commonly resulting in severe geographic tumor miss or necessitating excessively large isotropic margins (10-15 mm) that elevate normal tissue complication probabilities [3].

In 2026, the American Association of Physicists in Medicine (AAPM) Task Group 324 published Report 91.b ('The Management of Respiratory Motion in Radiation Oncology') [2], emphasizing that clinical ITV-to-PTV margins must be carefully calibrated to account for both residual respiratory irregularity and inter-fraction setup uncertainties, typically recommending margins of 4 to 5 mm. Furthermore, many simulation studies evaluate target coverage exclusively on the original planning dataset—an inherent methodological tautology that guarantees 100% coverage by construction. Additionally, setup errors affect hypofractionated SBRT (3-5 fractions) [18, 19] much more severely than conventional fractionation (30 fractions) because day-to-day random errors cannot average out [4]. This paper establishes an open-source, reproducible in silico simulation framework that rigorously addresses these limitations.

2. Materials and Methods

2.1 Anthropomorphic Thorax VOI and Tumor Geometry

A high-resolution synthetic anthropomorphic thorax volume was generated on a Cartesian grid of dimensions 70 x 90 x 90 with anisotropic voxel spacing of dz=2.0 mm, dy=1.5 mm, dx=1.5 mm (140 x 135 x 135 mm FOV). Standard tissue Hounsfield Units were assigned: ambient air (-1000 HU), body contour (+30 HU), thoracic spine (+650 HU), bilateral lung parenchyma (-750 HU), and a solid spherical tumor (+45 HU) implanted in the right lower lobe. Three lesion radii were evaluated: small nodule (r = 8.0 mm, V = 2.14 cc), nominal clinical tumor (r = 12.0 mm, V = 7.24 cc), and large lesion (r = 16.0 mm, V = 17.16 cc).

2.2 Asymmetric 60:40 Kinematics and Reference Phase

Standard 4D-CT acquisition bins 10 discrete phases from T0 to T9. Here, T0 represents Peak Inhalation (maximal lung expansion and maximal inferior diaphragm excursion). Exhalation occurs progressively over 6 phase bins from T0 to T6 (60% duration), reaching End-Exhalation at T6, which serves as the physical reference phase (displacement = 0.0 mm). Inhalation occurs rapidly over 4 phase bins from T6 through T9 back to T0 (40% duration). A modified Lujan power model (n=2) is employed with 36-degree AP/LR phase-lag hysteresis [5, 6].

2.3 Target Strategies and Geometric Surrogate Criteria

Target strategies include: (1) Isotropic PTV (2-12 mm around T6 GTV), (2) ITV + Setup Margins (0, 2, 3, 4, 5 mm, including TG-324 4-5 mm), and (3) Anisotropic Mid-position PTV (2, 3, 4 mm setup) [9, 10]. A study-defined geometric threshold of Coverage Ratio (CR) >= 95% was adopted as a surrogate evaluation criterion, analogous to ICRU 83 dosimetric D95% >= 95% recommendations [7].

[Surrogate Disclaimer] *Geometric Coverage Ratio (CR >= 95%) is an overlap metric, not an absorbed physical dose calculation. PTV volume reduction is a geometric proxy; 4D dose accumulation is required for clinical confirmation.*

2.4 Independent Robustness & Multi-Regimen Setup Simulation

Frozen planning PTVs were evaluated across N=5,000 independent unseen cycles (Monte Carlo with exact 95% Wilson score CIs) and a multi-regimen population setup uncertainty simulation across 50 virtual patients for SBRT 3-fx, SBRT 5-fx, and CFRT 30-fx regimens (total 1,900 sessions) incorporating independent systematic (Sigma=2.0 mm) and random (sigma=2.0 mm) errors: r_tumor(t, i, f) = r_resp(t) + S_i + R_{i,f} [4].

3. Results

3.1 Kinematics, Tracking Fidelity, and van Herk Margins

Table 1 reports kinematic parameters, tracking fidelity, and analytical van Herk margin recipes. Regular respiration produced excursions of LR = 1.20 mm, AP = 3.50 mm, and SI = 12.00 mm (Max 3D = 12.37 mm). Automated tracking achieved 3D MAE of 0.058 mm, 0.034 mm, and 0.021 mm for 8 mm, 12 mm, and 16 mm tumors, respectively.

Profile / Lesion Size	LR p2p	AP p2p	SI p2p	Max 3D	3D MAE	van Herk [LR,AP,SI]
Regular (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Deep Respiration	2.00 mm	6.00 mm	22.00 mm	22.59 mm	0.031 mm	[6.5, 7.1, 10.8] mm
Irregular Respiration	1.66 mm	4.38 mm	14.30 mm	14.74 mm	0.031 mm	[6.5, 6.8, 8.9] mm
Small Lesion (r=8mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.058 mm	[6.4, 6.7, 8.4] mm
Nominal Lesion (r=12mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.034 mm	[6.4, 6.7, 8.4] mm
Large Lesion (r=16mm)	1.20 mm	3.50 mm	12.00 mm	12.37 mm	0.021 mm	[6.4, 6.7, 8.4] mm

Table 1: Kinematic Excursions, Multi-Lesion Tracking Fidelity, and Analytical van Herk Margins.

Figure 1: Synthetic Anthropomorphic Thorax Phantom — Tissue HU Distribution

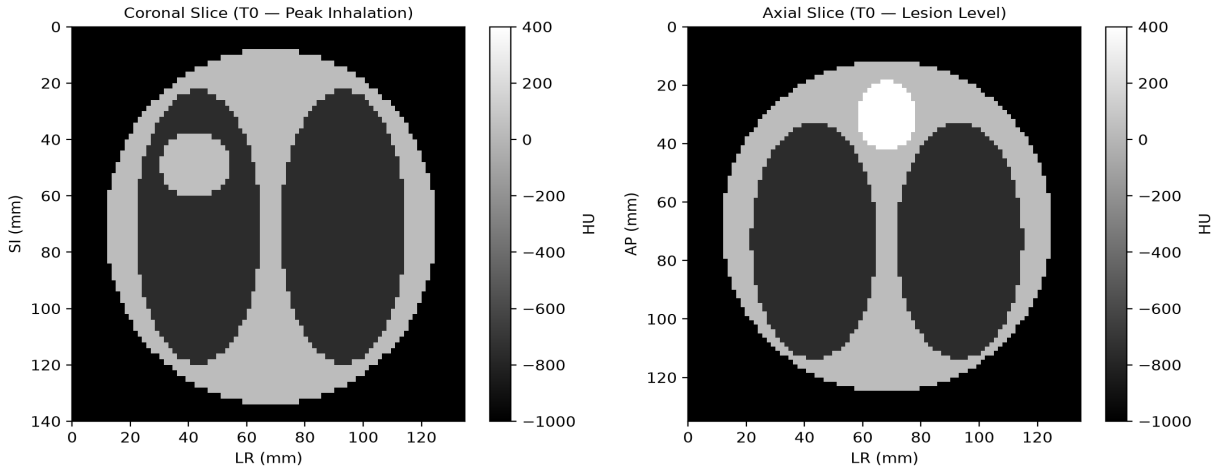


Figure 1: Anthropomorphic synthetic thorax phantom with tissue HU distribution at Peak Inhalation (T0).

Figure 2: 3D Tumor Centroid Trajectories (Reference: T6 End-Exhalation, 60:40 Cycle)

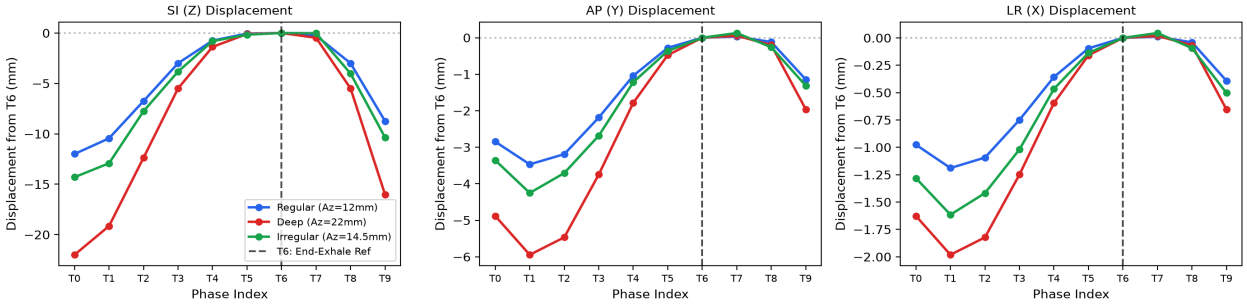


Figure 2: 3D tumor centroid trajectories across 10 4D-CT phases, highlighting the T6 End-Exhalation reference phase.

3.2 Isotropic Expansion Benchmark (Table 2)

Table 2 reports isotropic expansions from T6 GTV. Margins of 2 mm, 3 mm, and 5 mm achieved planning minimum coverage of 42.2%, 50.1%, and 64.7%, respectively. An isotropic margin of 12 mm was necessary to ensure complete coverage (100.0%) across all phases.

Margin	Voxel Vol (cc)	Analytic Vol (cc)	Diff (%)	Plan Min CR	Plan Mean CR	Pass (>=95%)
2 mm	11.3	11.5	-1.6%	42.2%	81.5%	Fail
3 mm	14.2	14.1	+0.5%	50.1%	85.8%	Fail
5 mm	20.6	20.6	+0.1%	64.7%	91.6%	Fail
8 mm	33.3	33.5	-0.7%	84.4%	97.3%	Fail
10 mm	44.3	44.6	-0.6%	94.5%	99.3%	Fail
12 mm	58.0	57.9	+0.2%	100.0%	100.0%	Pass

Table 2: Isotropic Expansion Benchmark: Volume, Discretization Fidelity, and Planning Coverage.

3.3 Target Strategy Volumes, Planning Coverage, and Multi-Regimen Robustness

Table 3 compares target strategies across planning coverage, Monte Carlo robustness (N=5,000), and multi-regimen setup uncertainty. ITV + 3 mm achieved 100.0% planning coverage (23.9 cc, +16.0% vs. baseline). Anisotropic PTV (Mid-pos + 2 mm) achieved 99.4% minimum coverage (19.3 cc, -6.3% vs. 5 mm baseline). In SBRT 5-fraction courses, ITV + 3 mm yielded only 12.0% all-fraction success, while ITV + 5 mm maintained 62.0% all-fraction success (and 82.0% in 3-fraction SBRT).

Strategy	Vol (cc)	vs 5mm	Plan MinCR	Irreg Pass (Wilson 95% CI)	SBRT 3fx AllPass	SBRT 5fx AllPass	CFRT 30fx FracPass
Isotropic 5mm	20.6	Baseline	64.7%	0.0% [0.0, 0.1]	0.0%	0.0%	1.9%
Isotropic 12mm	58.0	+181.6%	100.0%	29.9% [28.6, 31.1]	40.0%	40.0%	69.6%
ITV + 2mm setup	19.9	-3.5%	100.0%	55.4% [54.0, 56.8]	8.0%	2.0%	35.3%
ITV + 3mm setup	23.9	+16.0%	100.0%	87.1% [86.1, 88.0]	42.0%	12.0%	59.1%
ITV + 4mm (TG-324)	28.0	+36.0%	100.0%	87.2% [86.3, 88.1]	62.0%	32.0%	74.9%
ITV + 5mm (TG-324)	33.1	+60.6%	100.0%	97.9% [97.4, 98.2]	82.0%	62.0%	87.4%
Anisotropic (Mid-pos + 2mm setup)	19.3	-6.5%	99.4%	18.3% [17.3, 19.4]	0.0%	0.0%	19.9%
Anisotropic (Mid-pos + 3mm setup)	23.0	+11.6%	100.0%	53.1% [51.7, 54.5]	14.0%	2.0%	43.3%

Table 3: Target Strategy Comparison: Volume, Planning Coverage, and Multi-Regimen Robustness.

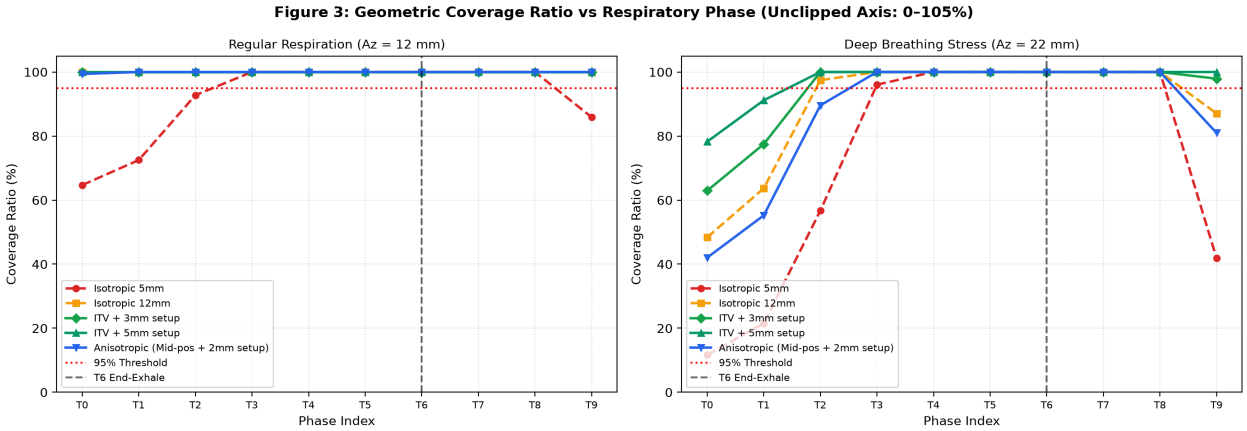


Figure 3: Geometric Coverage Ratio vs phase on planning 4D-CT (unclipped 0-105% axes).

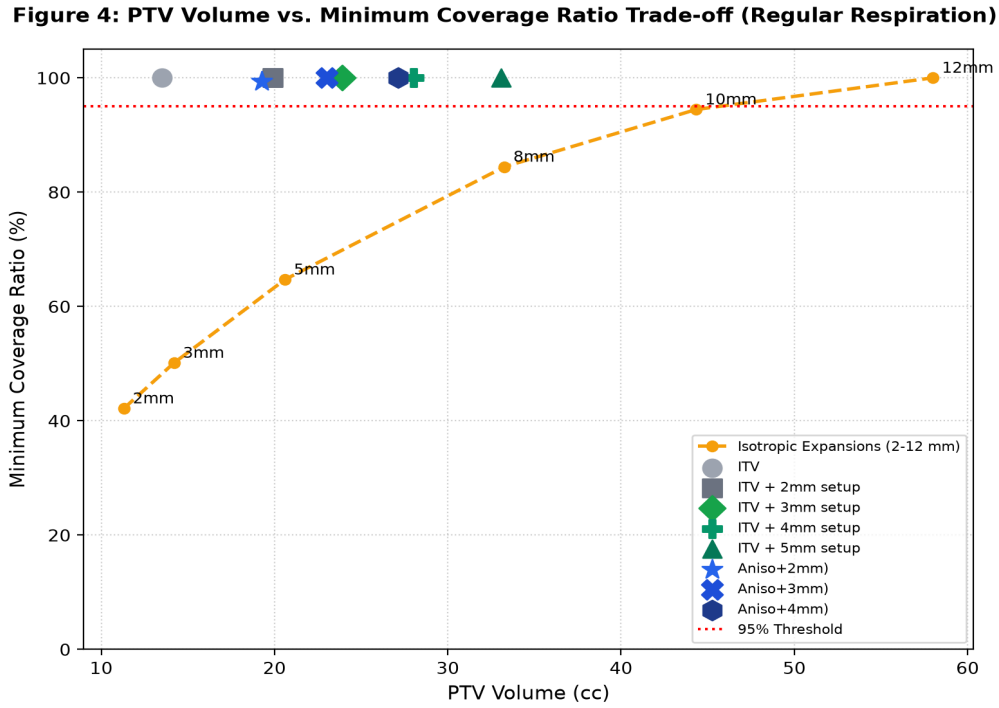


Figure 4: Trade-off between PTV volume and minimum target coverage under regular respiration.

3.4 Monte Carlo Robustness and Multi-Regimen Setup Uncertainty

Under irregular breathing with baseline drift (MC N=5,000), ITV + 3 mm pass rate dropped to 87.1% (95% Wilson CI: [86.1%, 88.0%]; mean min CR: 97.6%, empirical range: [86.5%, 100.0%]). Conversely, ITV + 5 mm maintained a 97.9% pass rate (95% Wilson CI: [97.4%, 98.2%]). In 5-fraction SBRT with 2 mm systematic and 2 mm random errors, ITV + 3 mm achieved only 12.0% all-fraction patient success, while ITV + 5 mm maintained 62.0% (and 82.0% in 3-fraction SBRT).

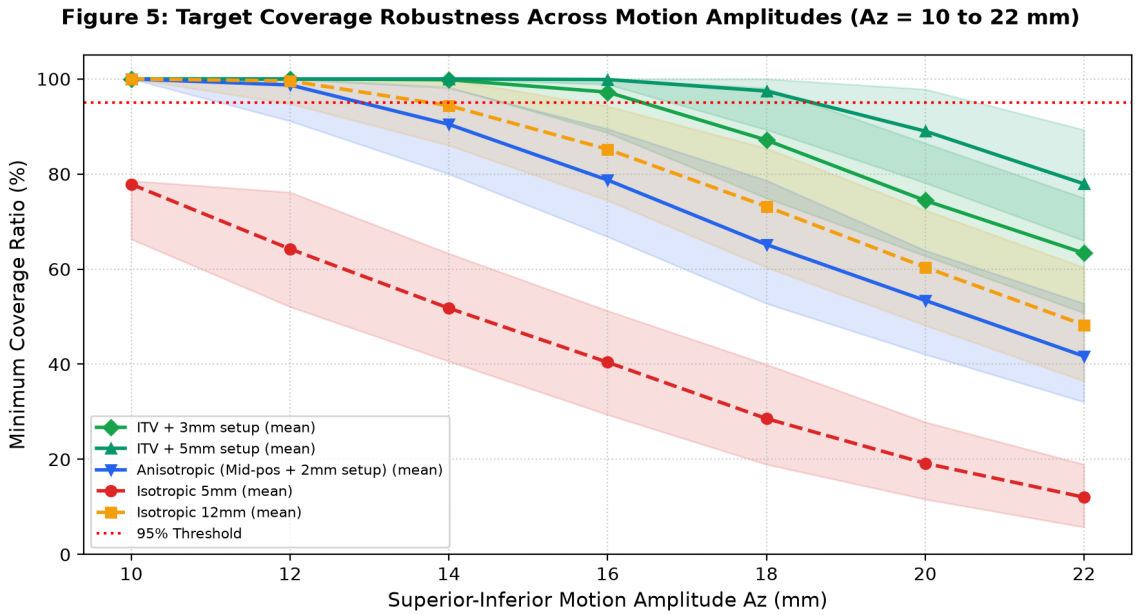


Figure 5: Target coverage robustness across motion amplitudes (Az = 10 to 22 mm) with 95% Wilson score CIs.

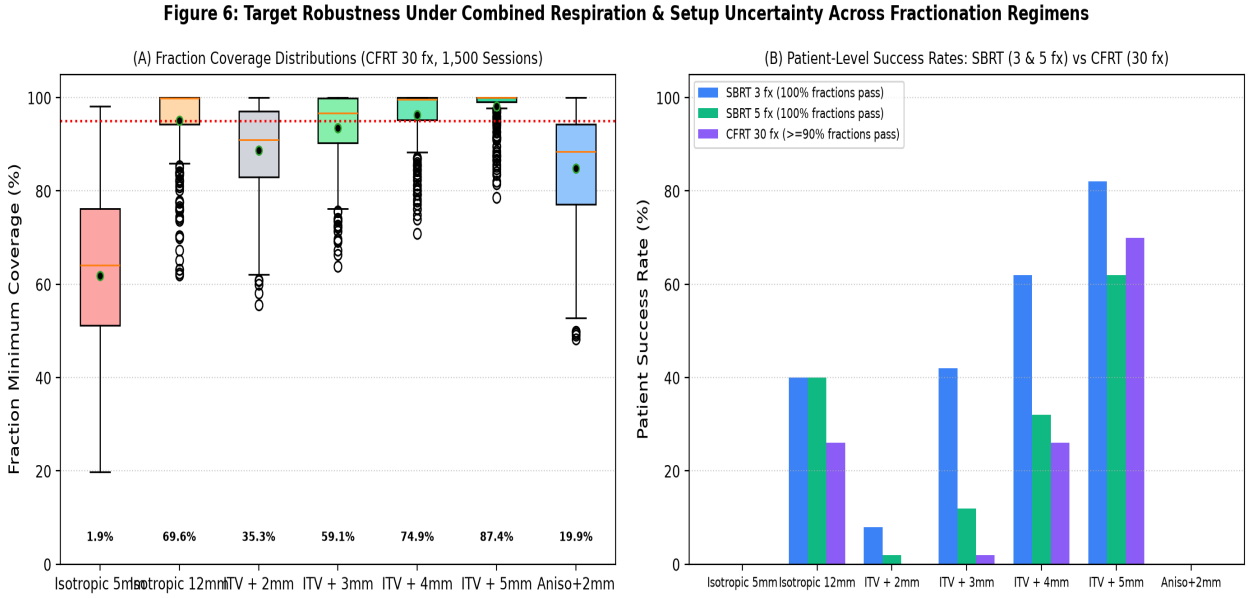


Figure 6: Setup uncertainty evaluation comparing CFRT 30-fraction distribution (left) and SBRT 3-fx & 5-fx all-pass rates (right).

4. Discussion and Limitations

This study demonstrates that while anisotropic and tight ITV margins (<= 3 mm) offer substantial theoretical normal tissue sparing under idealized regular breathing, they are highly vulnerable to baseline drift and setup uncertainty, particularly in hypofractionated SBRT. Increasing the ITV setup margin to 4-5 mm restores substantial coverage robustness, directly supporting AAPM TG-324 Report 91.b [2].

Study Limitations: (1) Rigid phantom target does not model non-rigid lung deformation [13]; (2) Coverage Ratio >= 95% is a geometric surrogate, whereas absorbed dose involves tissue density heterogeneities [5, 7]; (3) Real-world external surrogates can exhibit phase mismatch with internal motion [1, 12]; (4) Hypofractionated radiobiology involves biological effective dose escalation [18, 19].

5. Conclusion

While expanding to ITV + 5 mm substantially improves coverage robustness compared to tight 3 mm margins, geometric undercoverage persists in 12.6% to 38.0% of treatment sessions without daily correction. Therefore, in concordance with AAPM TG-324 Report 91.b, ITV margins of 4 to 5 mm are clinically required for free-breathing lung SBRT, but remain insufficient for complete margin reduction unless coupled with daily volumetric image guidance (CBCT) and active respiratory motion management.

Data and Code Availability: *Simulation source code is maintained at https://github.com/anhvu7123/4d-ct-lung-motion under the MIT open-source license.*

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